Software Requirements Specification

for

Qwik

Version 1.0

Prepared by

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Course: CS253

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

1.1 Product Scope

The goal of the project-Qwik is to digitalize the food ordering process in IITK. On the web platform, user will have a menu of all the campus canteens, allowing him/her to order online. This will save the user time, and they can see all the varieties available across different canteens. They can pay online. Also, one can also book a table or order a cake for any treats significantly ahead of time.

The first and foremost objective is to minimize the waiting time for the order placed which can be too much on occasion or busy hours, so Qwik can be the solution to this problem. And most find it hard to order for a treat or book a table so now they can simply do it online without physically present.

Qwik makes this entire process streamlined and easy for the canteen admin to manage orders. There will be two interfaces while one creates his/her account on the Qwik. One is for the user who are mainly students also for visitors called user account. The other is for canteen administrators called admin account.

The user while ordering first selects the Hall of Residence where he/she is going to receive their order which is set to the default Hall where he/she resides. After selecting items from the menu, the request is sent to the respective canteen admin where the admin will accept the items which are available then the user can edit and send a request again or place the order by paying via online. Once the order is ready the user will receive an alert saying the order is ready.

1.2 Intended Audience and Document Overview

Intended Audience

This document is intended for developers, project managers, testers, and other stakeholders involved in the project. It serves as a comprehensive guide, detailing the functional and non-functional requirements, system constraints, and implementation considerations to ensure successful development and deployment.

Document Overview

- **Introduction:** This section provides essential information to help readers understand the SRS, such as document conventions and abbreviations. If readers are already familiar with the basic terms, they may choose to skip this part. However, it serves as a handy reference to clear up any confusion while reading the document.

- **Overview of the system:** This part offers a general description of the software system, including its functionalities, assumptions, and dependencies. It is ideal for those seeking a quick understanding of the system and forms a good foundation for the more detailed sections that follow.

- **Specific requirements:** Here, you'll find an in-depth explanation of the software's features and functions, often supported by tree diagrams. This section is invaluable for end-users,

  - clients, and developers, serving as a guide during development and a user manual for future use.

- **Non-Functional requirements:** This section highlights the important non-functional requirements of the software. It holds particular importance for developers, guiding them in ensuring the system's performance, reliability, and other essential aspects.

- **Other Requirements:** This section captures essential requirements that do not fall under functional or non-functional categories but are critical for the system's success. It includes compliance, compatibility, security, documentation, and integration needs. This section ensures all additional considerations are documented to guide development and meet stakeholder expectations.

Appendix: This section provides supplementary information to support the main content of the SRS document. It includes the Data Dictionary, which defines the structure, format, and meaning of all data elements used in the databases, and the Group Log, which records key decisions, changes, and discussions among stakeholders during the project. This section serves as a reference for developers, testers, and other stakeholders to ensure clarity and consistency throughout the project lifecycle.

1.3 Definitions, Acronyms and Abbreviations

- **UPI:** Unified Payments Interface, used for cashless transactions.
- **UI:** User Interface. The part of the system with which the user interacts.
- **DBMS**: Database Management System. Software used to store, retrieve, and manage data in databases for the Qwik system.
- **API:** Application Programming Interface – Set of protocols for communication between different parts of the Qwik system or with external systems.
- **OTP:** One-Time Password. Used to verify Guest Users who signed up using their phone numbers.
- **SMS:** Short Message Service. OTPs are sent to Guest Users via SMS.
- **Website:** A comprehensive functional and non-functional system that must be clearly described for development.
- **Dashboard:** A personalized interface showing order history, ongoing bookings, and account details.
- **Live Updates:** Real-time information provided to users about their orders, including preparation, estimated delivery time, and completion status.
- **Peak Periods:** High-demand times during which the canteen experiences a surge in orders.
- **Event Booking:** A feature in Qwik that allows users to reserve space and arrange catering for special events on campus.

1.4 Document Conventions

- **This document follows the IEEE standards for Software Requirements Specification (SRS).**
- The document is written with the **Arial** font of size 11 with single spacing and 1 Inch margins.
- Important keywords are highlighted in bold font and comments are italicized
- Section titles are hierarchical and numbered, with clear and concise descriptions under each heading.

1.5 References and Acknowledgments

We express our profound gratitude to our TA, Utkarsh Agarwal, and our course instructor, Professor, Indranil Saha, for their valuable guidance throughout the creation of this document and for providing us with a template for the Software Requirements Specification.

References include relevant course materials and initial project discussions.

2 Overall Description

2.1 Product Overview

Qwik is an interactive food ordering system developed to simplify and improve the dining experience within the IITK campus. The platform focuses on providing easy access to campus canteens and eateries while making the meal ordering process faster, more efficient, and user-friendly.

The system is designed to support different categories of users, primarily customers and canteen managers. Canteen managers are provided with tools to handle incoming orders and reservations, update and control menu items, and analyse operational performance. On the other hand, customers can browse available menus, place orders, monitor order status in real time, review past orders, and manage their personal profiles.

To deliver a secure and personalized user experience, Quick Crave incorporates email-based account verification, supports multiple payment options, and offers tailored features for individual users. Additionally, guest users are given the flexibility to register using their mobile numbers, with authentication handled through an OTP-based SMS verification system.

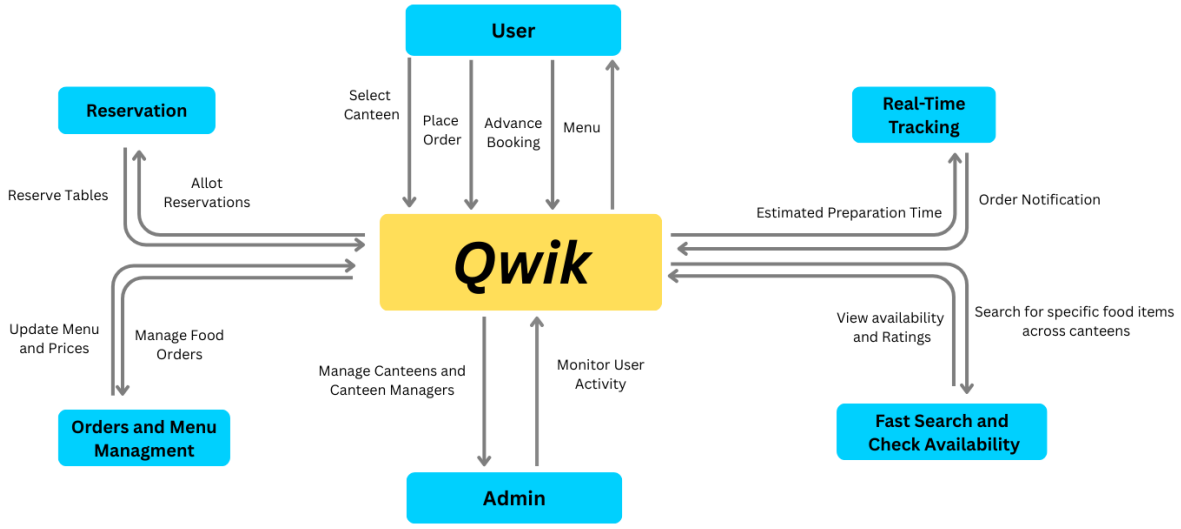


Figure 1 : Overview Flowchart

2.2 Product Functionality

2.2.1 User Interfaces

• **Account Creation and Role-Based Access:**
– Users can create accounts with email-based OTP verification.
– Role options include Customer, Canteen Manager, Admin.

• **Advanced Search and Availability:**
– The home page displays food items in a card-based layout, like Swiggy.
– Users can select a specific canteen and view all available food items from that canteen directly on the home screen.
– Users can also check the ratings on the food item card.

• **Seamless Order Placement and Real-Time Tracking**:
– Customers can submit orders, which are processed only after approval from the canteen manager.
– Users receive in-app notifications when their orders are ready for pickup.
– Users can track active orders along with updates on the estimated preparation time.

• **Advance Food Preordering:**
– Customers can prebook food items for specific time slots to ensure availability and timely service during peak hours.

• **Specialized Ordering and Event Reservations:**
– Users have the option to order custom-designed birthday cakes with their preferred flavors.
– Users can reserve canteen spaces for private events such as birthdays and small gatherings.

• **User Dashboard for Order Management:**
– Users can access and review their order history and view estimated preparation times through an intuitive user dashboard.

2.2.2 Canteen Manager Interfaces

• **Order and Menu Management:**

– Handle incoming orders, approve requests, and update order statuses.

– Manage reservation requests.

– Add, update, or remove menu items, including pricing and availability.

• **Availability Scheduling:**
– Define operational hours and set availability for specific days.

- **User Management and Moderation:**

. –Monitor user activity and order behavior.

–Restrict or block user accounts when necessary.

2.2.3 Admin Interfaces

- **Canteen Management:**
  - Add, Update or Remove canteens from the system.
  - Accept and manage accounts for canteen managers.
  - Approve, modify, or remove food items added by canteen managers.
  - An option will be provided to block users who misuse the ordering system or place unnecessary requests.
- **System Configuration and Maintenance:**
  - Enable or disable features globally.
  - Monitor user activity and order flow across canteens.

2.3 Design and Implementation Constraints

The development of Qwik is subject to the following constraints:

- The application shall be developed using HTML, CSS, JavaScript, and ReactJS for the frontend, and Node.js with Express.js for the backend.
- Data storage will be handled using MongoDB or PostgreSQL, depending on the data requirements.
- The system should be capable of supporting up to 2,000 concurrent users while maintaining a response time of under three seconds.
- The system will be available from time to time in accordance with canteen operating hours.
- Deployment will initially use AWS free-tier services, with scalability options in place for production.
- The system will support mobile platforms, with the mobile application available on Android devices.

2.4 Assumptions and Dependencies

The successful operation of Qwik relies on the following assumptions and dependencies:

- Users will have reliable internet access and provide accurate information.
- Canteen Manager will regularly update menus and food availability.
- The system depends on third-party services such as email notifications and payment gateways like UPI (PhonePe, GooglePay, PayTM etc)
- The system relies on server availability and internet connectivity for real-time updates and notifications.
- Compliance with IIT Kanpur's IT policies and data security guidelines is mandatory.
- The system will adhere to relevant food safety regulations and data protection laws.

3 Specific Requirements

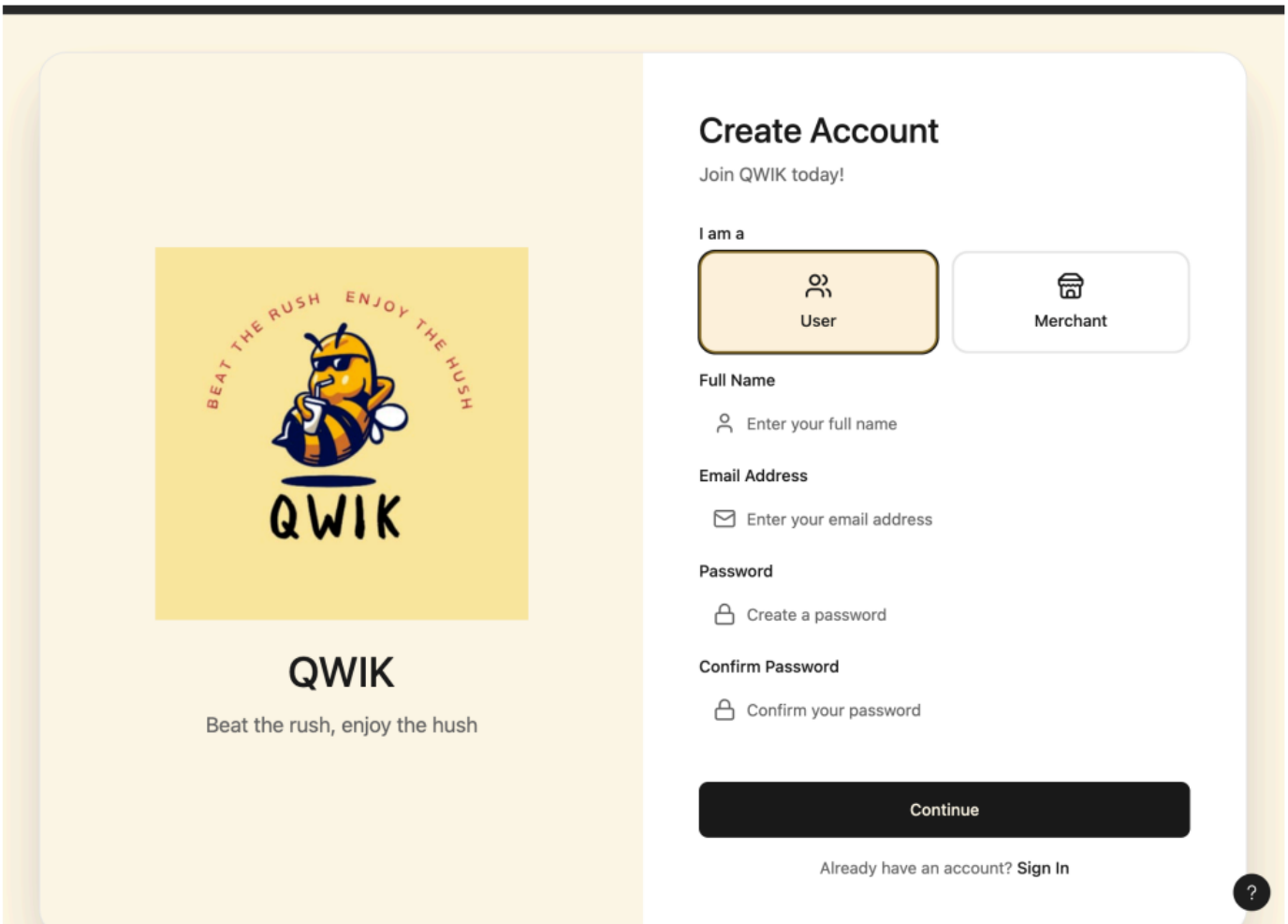
3.1 External Interface Requirements

3.1.1 User Interfaces

The Digital Canteen Management System provides a web-based graphical user interface (GUI) with role-based access, consisting of three primary user interfaces: the Student Interface, the Merchant Interface, the Super Admin Interface.

STUDENT INTERFACE (Customer Side)

CREATE ACCOUNT PAGE:



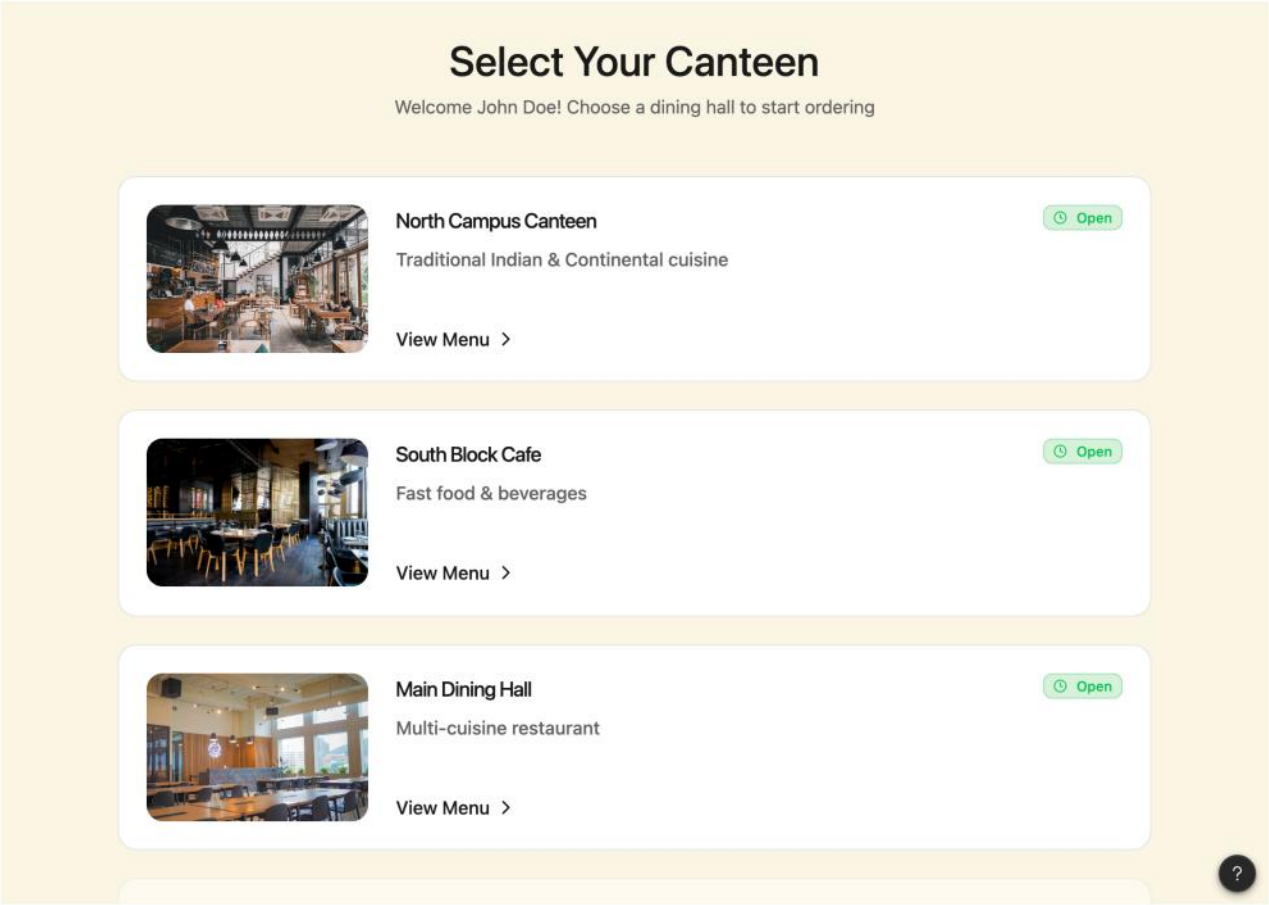
The Create Account or SIGN UP page allows new users to register on the website by providing the required personal details. Users can select their role as either a User or a Merchant. The page collects information such as full name, email address, and password. Users are required to create and confirm a password that follows predefined security rules. Upon successful

LOGIN PAGE:

The Login page allows registered users to access the system by entering their registered email address mobile along with their password. Users can authenticate themselves by selecting the Sign In/Login option.

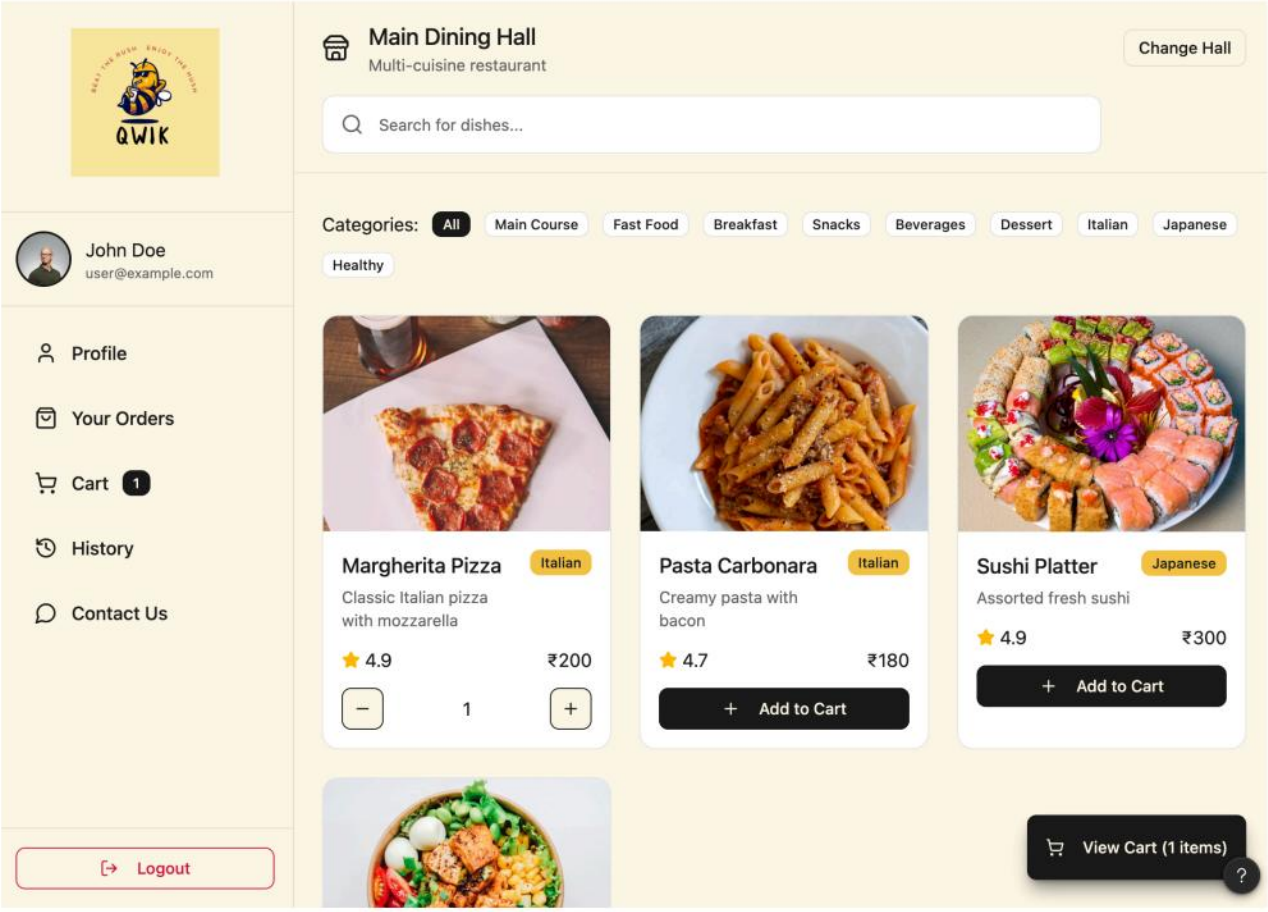
In case of forgotten credentials, the Forgot Password feature enables password recovery through OTP verification. Successful authentication redirects the user to the main application interface.

HALL SELECTION:



The Hall Selection page allows users to select a preferred dining hall from the list of available halls on campus. Each hall is displayed with its name and relevant details to assist in selection. Upon selecting a hall, the user is redirected to the corresponding food menu page. The selected hall is saved for the current session to personalise the ordering experience.

HOME PAGE:

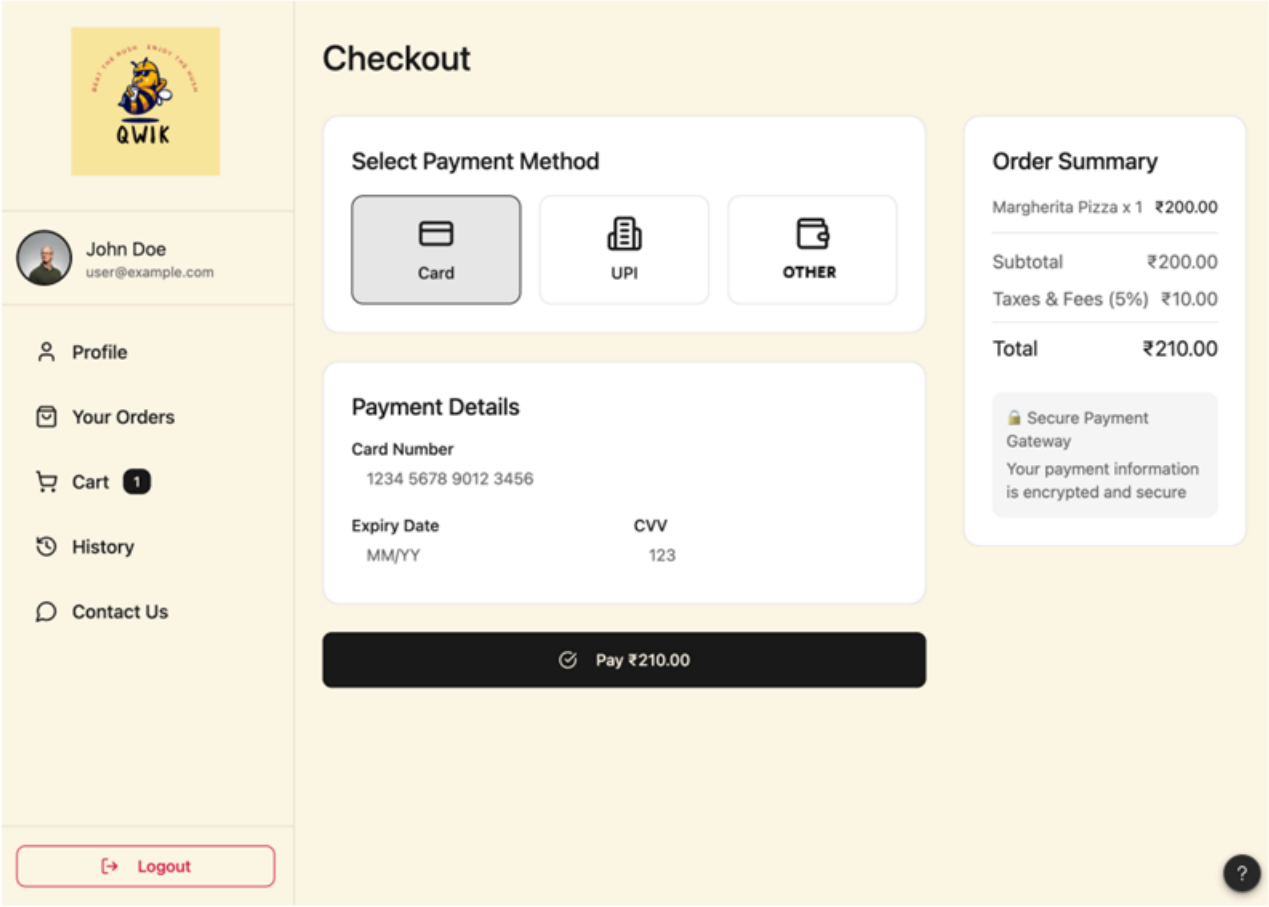


This Home page displays the list of available dishes offered by the selected dining hall (In the above example it is Main Dining Hall). Users can search for dishes using the search bar and filter items based on categories such as cuisine type or meal preferences. Each food item shows relevant details including name, description, price, and rating. Users can adjust item quantities and add items to the cart. The page also provides navigation options to view the cart, order history, profile, and other related sections. And we can also change canteen from the option Change Hall.

Your Orders - Shows the list of current placed by the user.
Cart -Allows users to view selected items and proceed to checkout.

History -Allows users to view all past orders placed by him.

CHECKOUT PAGE:



The Checkout page enables users to review their selected items and complete the payment process. Users can choose a preferred payment method such as Card, UPI and some other options. The page displays an order summary including item details, subtotal, applicable taxes, and the final payable amount. After, the order is accepted by merchant side Hall Manager, we can enter the required payment details, users can confirm the payment to successfully place the order. Secure payment processing is ensured during the transaction.

MERCHANT INTERFACE (Canteen Side)

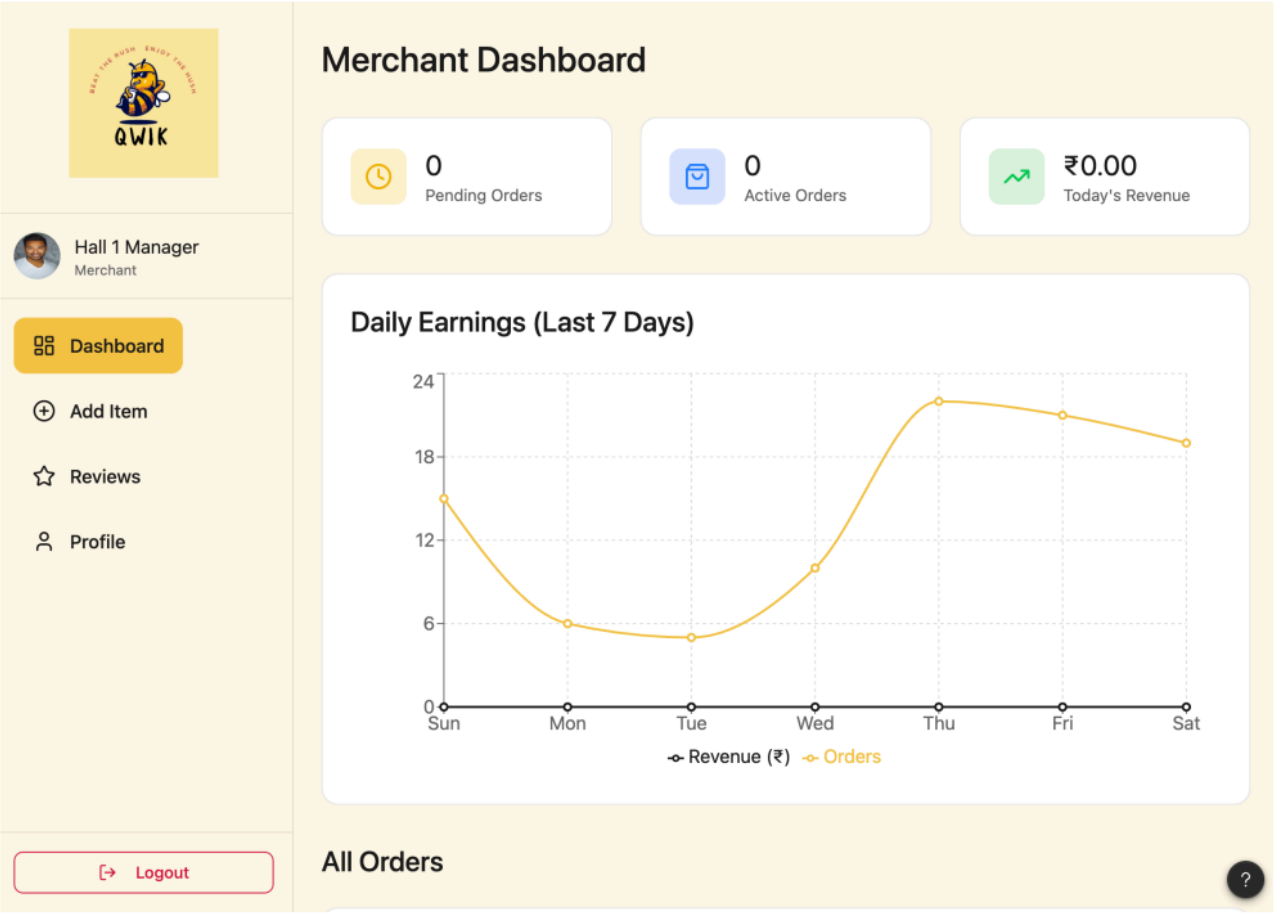
CREATE ACCOUNT PAGE:

The Merchant Create Account page allows merchants to register on the platform by providing required details such as name, email address, and password. Merchants can create and confirm their credentials to access the merchant interface. This page functions similarly to the user registration page with role-based account creation.

LOGIN PAGE:

Login page enables registered merchants to access the merchant portal by entering their credentials. Merchants can authenticate using their email and password to manage orders and food items. This page is similar to the user login page but redirects to the merchant dashboard after successful authentication.

MERCHANT DASHBOARD:



The Merchant Dashboard serves as the central control panel for merchants to manage their activities on the platform. It provides a summary of key operational metrics such as pending orders, active orders, and daily revenue.

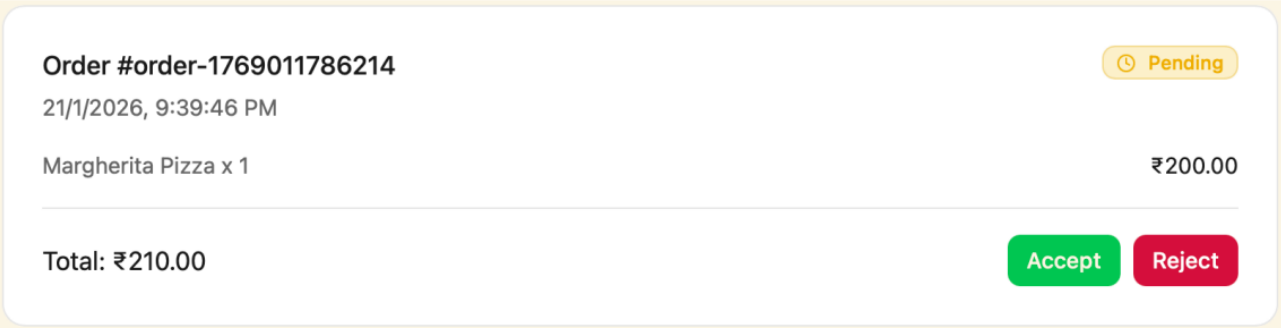
The dashboard also displays graphical insights on earnings and order trends over a selected time period to help merchants analyse performance. From this page, merchants can monitor order status, track business progress, and from their side bar they can navigate to other management features such as food item and orders.

In Sidebar:

Food Items -Allows merchants to add, edit, and manage menu items, including updating item details and availability status.

Orders - Enables merchants to view and manage current and past orders placed by users.

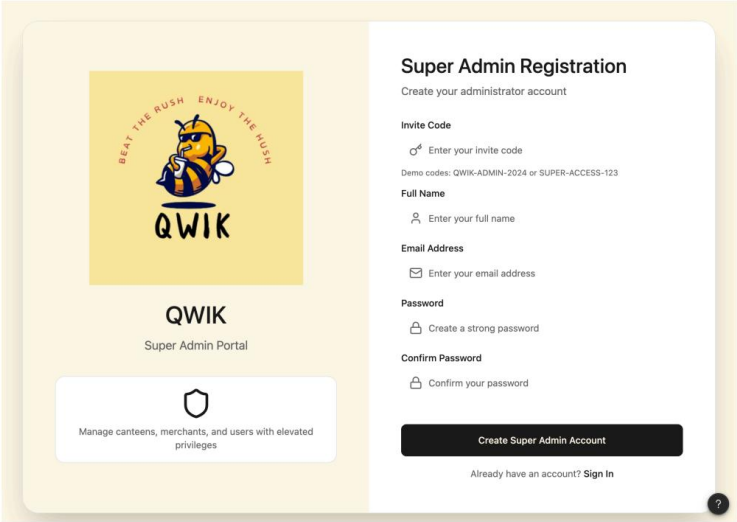
CURRENT ORDER DETAILS:



The Order Details page displays information related to an current order including order ID, timestamp, ordered items, and total amount. It allows the merchant to review the order status and take appropriate actions such as accepting or rejecting the order and updating Order status

SUPER ADMIN INTERFACE

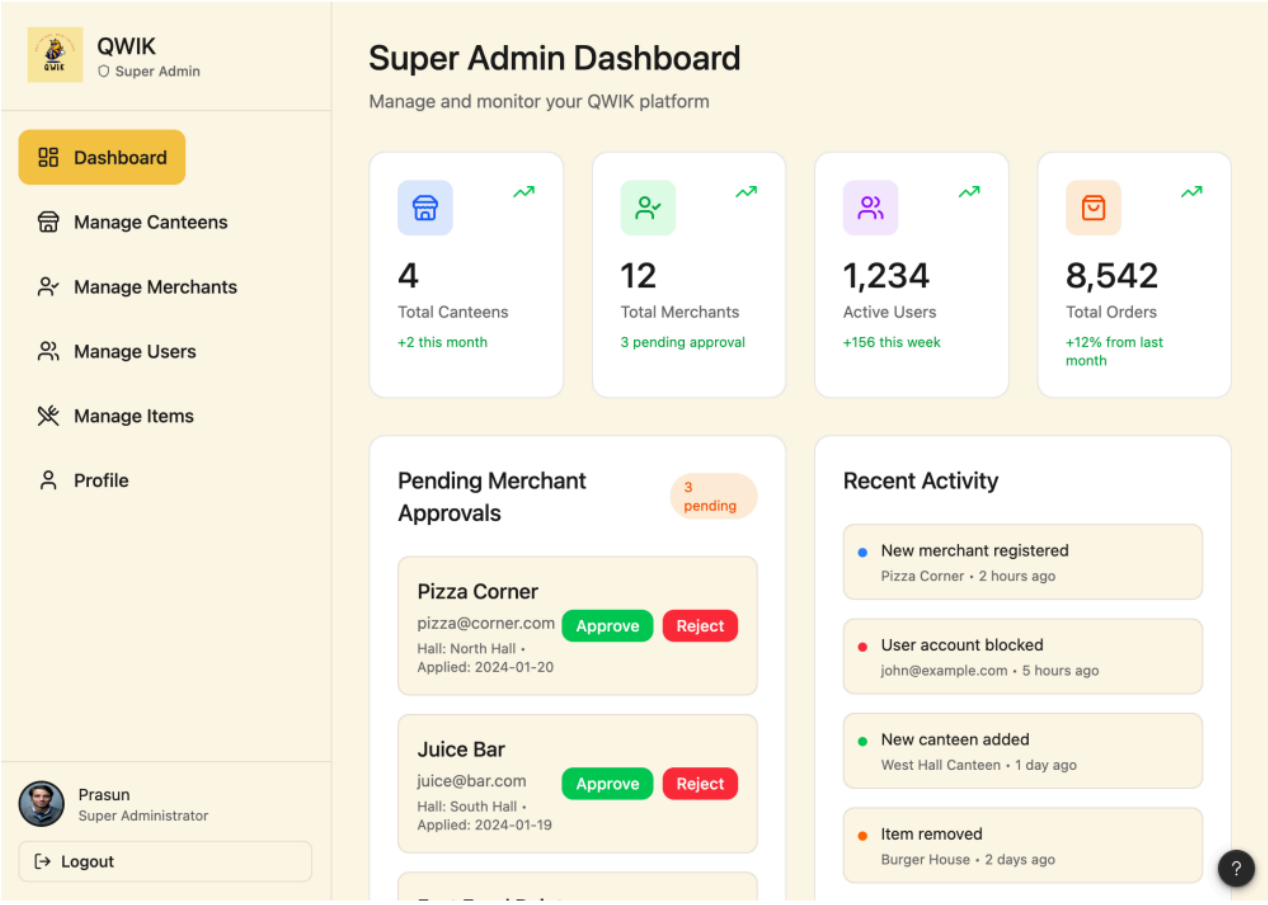
SUPER ADMIN REGISTRATION PAGE



The Super Admin Registration page allows authorized administrators to create a super admin account using a valid invite code. The page collects essential details such as full name, email address, and password.

Account creation is restricted through invite-code verification to ensure secure access. Upon successful registration, the administrator is granted elevated privileges to manage the platform.

SUPER ADMIN DASHBOARD



The Super Admin Dashboard serves as the central monitoring and control interface for the entire platform. It provides an overview of key system metrics including total canteens, merchants, active users, and total orders.

The dashboard displays pending merchant approval requests, enabling the administrator to approve or reject merchant applications. It also shows recent system activities such as user actions, canteen updates, and item modifications. This page allows the super admin to efficiently monitor platform performance and manage administrative operations.

3.1.2 Hardware Interfaces

The Digital Canteen Management System interacts with a limited set of hardware devices used by students and merchants to access the system and perform daily operations. These interfaces support authentication, order placement, order tracking, and order fulfilment at the canteen counters.
The following hardware components are required for system operation.

Student Devices

1. **Smartphones and Tablets**
   Students primarily access the system through mobile devices using a web browser or application. These devices are used to browse menus, place orders, receive status updates, and display order numbers at the counter.

2. **Personal Computers and Laptops**
   Students may also use desktop or laptop computers to access the platform for placing orders and tracking orders.

Merchant Devices

3. **Merchant Counter Terminals**
   Canteen operators use desktop computers, laptops, or tablets at the counter to access the merchant dashboard for viewing incoming orders, confirming orders, updating preparation status, and managing menu availability.

4. **Registered Mobile Phones**
   Merchants use mobile phones for OTP-based login authentication and for receiving system notifications.

Backend Hardware

5. **Central Server System**
   A centralized server hosts the application logic and databases required for user management, order processing, real-time synchronization, and review storage.

Software Interfaces

- The application is accessed through standard web browsers such as Chrome, Firefox, or Safari. The user interface is implemented using HTML, CSS, and JavaScript, enabling interaction between users and the system.
- The frontend communicates with the backend server through RESTful APIs. These APIs handle business logic, user authentication, authorization, and request processing.
- The backend server interfaces with a relational database MongoDB to store and retrieve user data, transaction records, and system configurations.

3.2 Functional Requirements

3.2.1 F1: Authentication and security

- A user shall be able to authenticate into the system using a valid email address, name and password.
- After entering the details, the system shall send a one-time password (OTP) to the registered email address for verification purposes.
- Upon successful registration the system shall uniquely identify each registered user, merchant, and super administrator using a system-generated unique ID and add their account to Databases.
- The system shall restrict access to features and data based on the authenticated user's role (user, merchant, or super administrator).
- The user can access his account any time after a successful registration using email and password in SIGN IN/Login page.
- The system shall allow users to securely reset their password using a verification mechanism.
- The system shall terminate user sessions upon logout to ensure secure access control.

3.2.2 F2: Customer Functionalities

ORDERING:

- After successful login, the user will be able to select the canteen from which they wish to place an order. Based on the selected canteen, the system will present the available food options to the user.
- The system will display the list of food items offered by the selected canteen along with relevant details such as item name, price, brief description, and rating.
- The user will be able to add required food items to the cart and modify the cart contents at any time before placing the order. This includes updating item quantities or removing items from the cart as needed.
- After every time user makes any changes happen in cart the system shall calculate the total order cost including applicable taxes and fees.
- The user will be able to check out the cart and proceed to place an order request.
- Upon placement of the order request, the system will forward the request to the respective merchant for confirmation.
- Upon completion of an order, the system shall update the order status and notify the user on the website.

PAYMENT:

- After an order request has been accepted by the respective merchant, the user will be redirected to the payment process to complete the transaction. The payment process will require the user to choose a preferred payment method from the available options such as UPI, card or other available options.
- During the payment process, the system will calculate and display the final

payable amount, including applicable taxes and additional charges, if any.

- Upon successful payment, the system will confirm the order and update its status accordingly.
- Both the user and the merchant will receive a notification indicating successful payment and order confirmation.
- In case of a payment failure, the system will notify the user and provide an option to retry the payment
-

PARTY MODE:

- The Party Mode allows users to pre-order and customise cakes for special occasions by selecting a preferred date and time for pickup. This enables users to plan celebrations in advance.
- After the preorder request the system will send it to the respective merchant for confirmation and preparation.
- users will be able to reserve tables by providing time, date, and an approximate number of people attending the event. This information will assist the merchant in managing seating arrangements.

3.2.3 F3: Merchant Functionalities

- Menu Management:

  The system provides options for the canteen manager to manage the menu by adding new food items, editing existing item details, updating prices, uploading item images, and modifying item availability based on stock or operational conditions.

- Order Notifications:

  The system notifies the canteen manager whenever a new order request is received, enabling quick access to the order details for further action.

- Order Queue Management:

  The canteen manager can view incoming orders and manage the order queue by accepting or rejecting orders. Accepted orders can be marked as under preparation to reflect real-time order status.

- Party Mode Management:

  The system allows the canteen manager to view and manage cake pre-order requests received through Party Mode, including reviewing customization details and scheduled pickup times.

- Customer Communication:

  The system provides an option for the canteen manager to notify customers regarding order updates or important information related to their orders.

- Statistics and Analytics:

  The system displays summary statistics such as active orders, today's orders, daily earnings, and order trends to help the canteen manager analyze business performance.

- Canteen Availability Management:
  The canteen manager can manage canteen operating status by marking the canteen as open or closed to users and specifying unavailable days when required.

3.2.4 F4: Account Management

- A user shall be able to view their account profile information from profile option in Navigation bar,
- Users will be able to update their personal information such as name, email address and profile details. Any changes made to sensitive information may require verification and system will update the new details to databases.
- The system provides users with the option to change their account password to maintain security and privacy.

3.2.5 F5: Super-Admin Functionalities

- The system ensures that super admin will be able to create an account only through an authorized invite link provided by the system developers.
- The super admin will be provided with access to an administrative dashboard that presents an overview of the entire system. This includes information related to users, merchants, canteens, orders, and overall platform activity.
- The super admin will be able to review and process merchant registration requests.
- The system allows the super administrator to approve or reject merchant applications, ensuring that only authorized merchants are permitted to operate on the platform
- The super admin will be able to manage user accounts by viewing user details and taking necessary actions such as blocking or unblocking accounts in case of policy violations or misuse.
- the system shall allow the super administrator to monitor food items and related activities across all canteens.
- The system ensures that all changes made by the super administrator are validated, securely applied, and reflected across the platform in real time

3.3 Customer Use Case Model

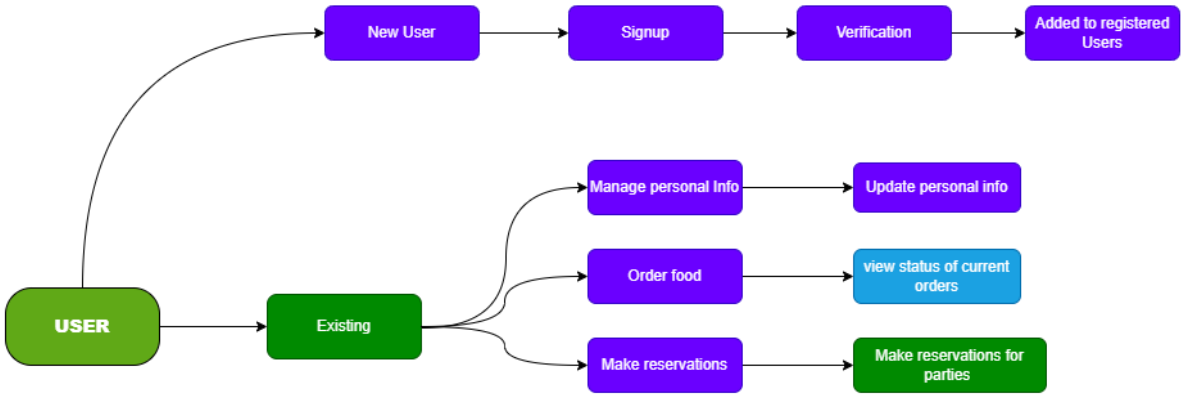


Figure 1: Customer Use Case Model

3.3.1 Customer Use Case #1: Food Ordering

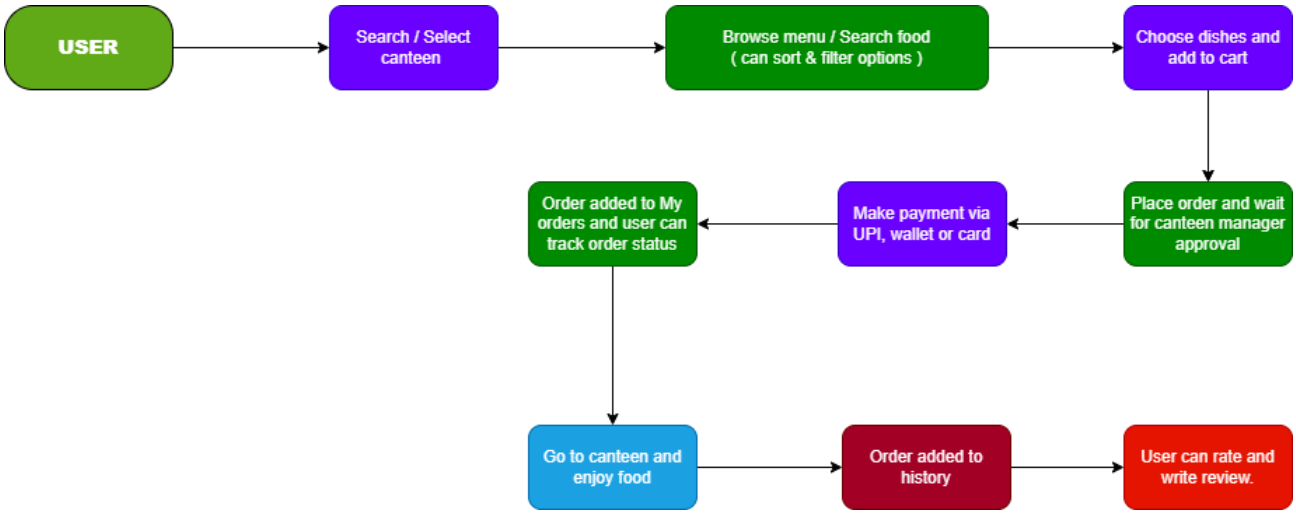


Figure 2: Customer Use Case 1

Author – Katroth chandrakanth

Purpose - The purpose of this use case is to allow a user to search for a canteen, browse food items, place an order, make payment, and track the order until it is completed. It helps users order food easily without standing in a queue

Requirements Traceability –

- User can search and select a canteen
- User can view menu items
- User can add food items to cart

- User can make online payment
- User can place order and track order status
- User can view order history
- User can give rating and review

Priority – **High**
This is the main functionality of the system and must work properly when the system is deployed.

Preconditions –

- User must be registered and logged into the system
- Canteen and food menu must be available in the system
- Internet connection should be available

Post conditions –

- Order is successfully placed
- Payment is completed
- Order is added to order history
- User can track order status
- User can rate and review after order completion

Actors –

- **Primary Actor:** User (Customer)
- **Secondary Actors:**
    - Payment Gateway (UPI / Card)
    - Canteen Manager
    - System (Food Ordering Application)

Exceptions –

- Payment failure
- Selected food item becomes unavailable
- Internet connection failure
- Canteen manager rejects the order
- System error while placing order

Includes (other use case IDs): 3.4.1 - Manager should accept the orders

Notes/Issues –

- Order status should be updated in real time
- User should receive notification after order confirmation

3.3.2 Customer Use Case #2: Making a Reservation

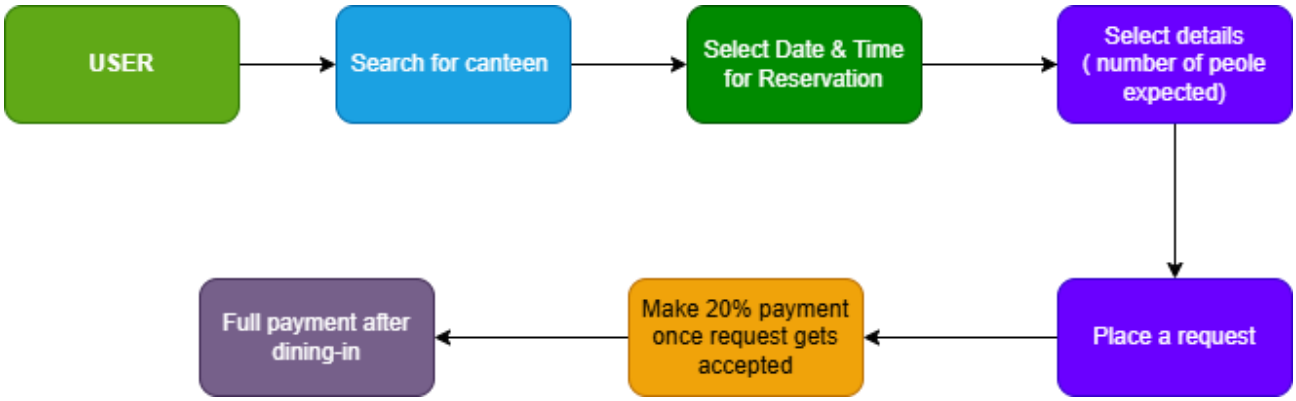


Figure 3: Customer Use Case 2

Author – Ramalingam Naidu Allu

Purpose - The purpose of this use case is to allow the user to reserve a table in a canteen by selecting date, time, and number of people. The user makes a partial payment after the reservation request is accepted and completes the full payment after dining.

Requirements Traceability –

- User can search for a canteen
- User can select reservation date and time
- User can enter number of people
- User can place a reservation request
- User can make partial payment (20%)
- User can make full payment after dining

Priority – Medium
This feature is important for planning visits but is not as critical as food ordering.

Preconditions –

- User must be logged into the system
- Canteen must support table reservation
- Available time slots should exist
- Internet connection must be available

Post conditions –

- Reservation request is successfully placed
- Reservation is confirmed by canteen
- Partial payment (20%) is completed
- Full payment is completed after dining

Actors –

- **Primary Actor:** User (Customer)
- **Secondary Actors:**
  - Canteen Manager
  - Payment Gateway
  - System (Reservation Module)

Exceptions –

- Selected time slot is not available
- Reservation request is rejected
- Payment failure during advance payment
- User does not show up for reservation
- System or network error

Includes (other use case IDs): 3.4.1 - Manager should accept the orders

Notes/Issues –

- Cancellation and refund rules should be clearly defined
- Advance payment percentage can be changed by admin
- User should receive confirmation notification

3.3.3 Customer Use Case #3: Account Management

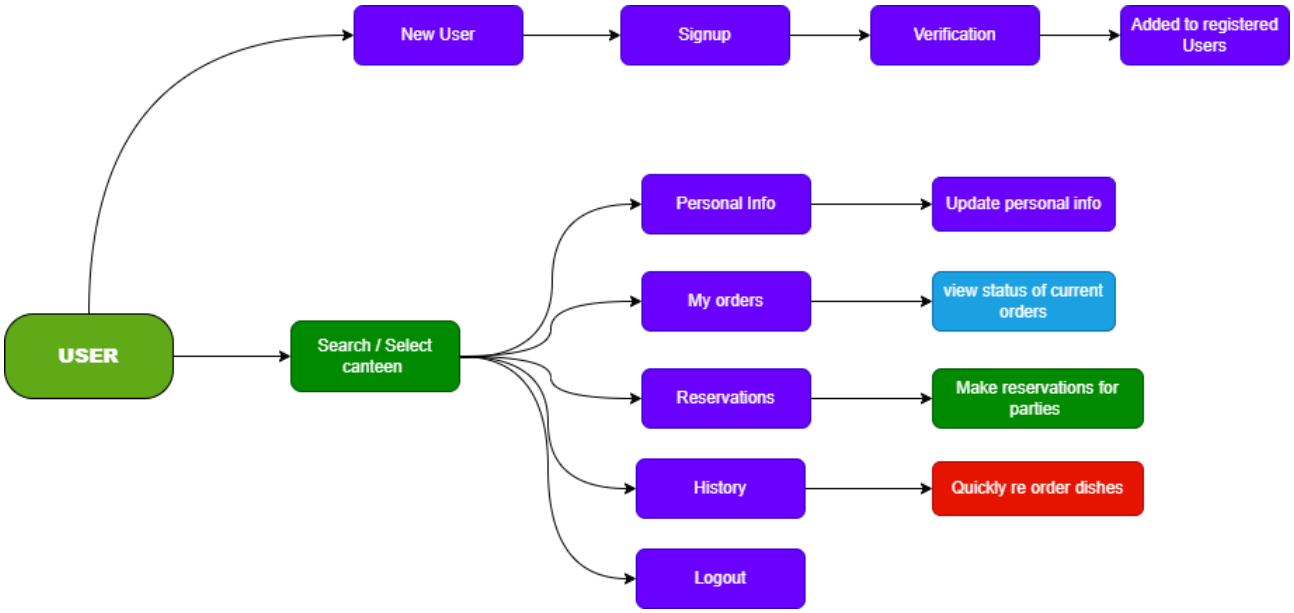


Figure 4: Customer Use Case 3

Author – Kallu Venkata Prasun

Purpose - The purpose of this use case is to allow a user to register, manage personal details, view orders and reservations, and use account features like reordering food and logging out.

Requirements Traceability –

- New user can sign up
- User can verify account
- User can update personal information
- User can view current order status
- User can view past orders and history
- User can make reservations for parties
- User can quickly reorder dishes
- User can log out securely

Priority – High
User account management is important for using all other features of the system.

Preconditions –

- User must have access to the application
- Internet connection must be available
- User must complete verification for full access

Post conditions –

- User account is created successfully
- Personal information is saved or updated
- User can access orders, reservations, and history
- User is logged out safely when logout is selected

Actors –

- **Primary Actor:** User
- **Secondary Actors:**
  - System (Authentication and User Management Module)

Exceptions –

- Signup or verification failure
- Incorrect login details
- Unable to update personal information
- System error while loading user data

Includes (other use case IDs): This is an Independent Case

Notes/Issues –

- Password should be stored securely
- User session should expire after logout
- Account verification should be completed before placing orders

3.4 Manager Use Case Model

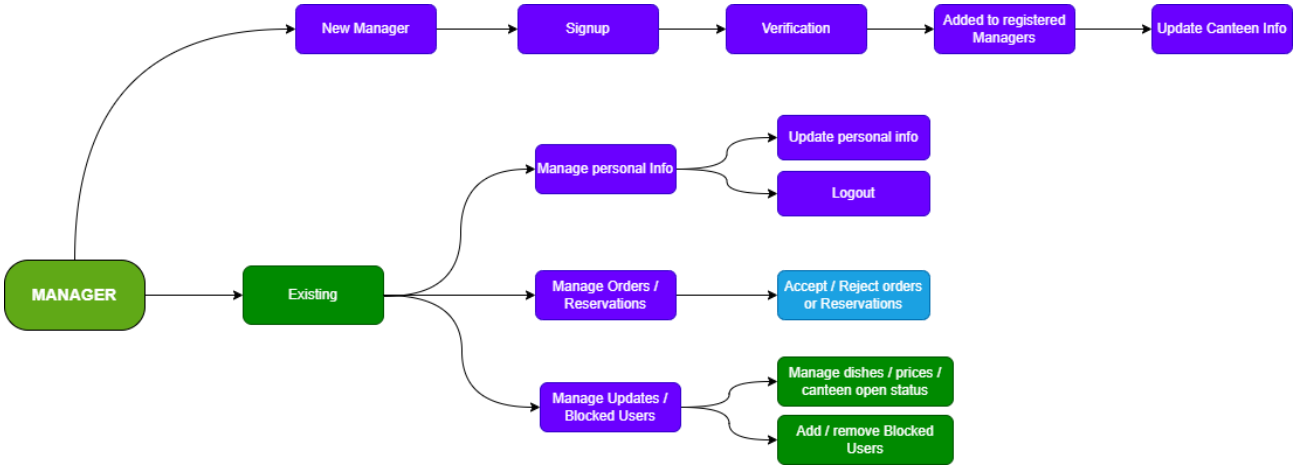


Figure 5: Manager Use Case Model

3.4.1 Manager Use Case #1: Food Ordering

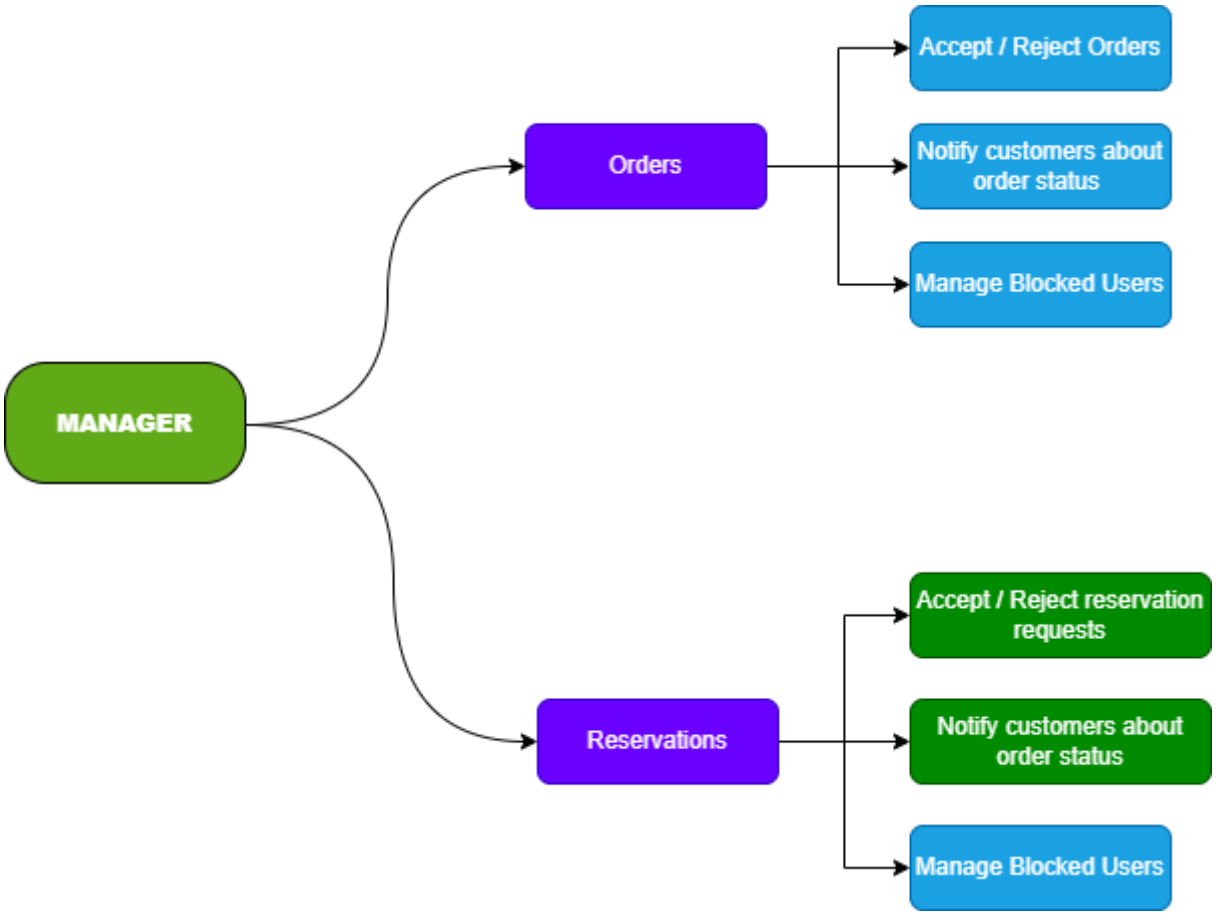


Figure 6: Manager Use Case 1

Author – Sumanth Reddy Kamjula

Purpose - The purpose of this use case is to allow the canteen manager to receive food orders from users, approve or reject orders, manage order status, and ensure food is prepared and delivered on time.

Requirements Traceability –

- Manager can view incoming food orders
- Manager can approve or reject orders
- Manager can update order status
- Manager can manage menu availability
- Manager can view order history
- Manager can view customer ratings and reviews

Priority – High
This use case is critical because orders cannot be completed without manager approval and processing.

Preconditions –

- Manager must be logged into the system
- Orders must be placed by users
- Menu items must be available in the system
- Internet connection must be available

Post conditions –

- Order is approved or rejected
- Order status is updated in the system
- User is notified about order status
- Order is completed and added to history

Actors –

- **Primary Actor:** Canteen Manager
- **Secondary Actors:**
    - System (Order Management Module)
    - User (receives order updates)

Exceptions –

- Order cannot be fulfilled due to item unavailability
- Manager rejects the order
- System error while updating order status
- Delay in order preparation

Includes (other use case IDs): 3.3.1 & 3.3.2 - User should order food

Notes/Issues –

- Manager should update order status in real time
- Rejected orders should trigger refund process
- High order volume may cause delays

3.4.2 Manager Use Case #2: Account Management & Operations

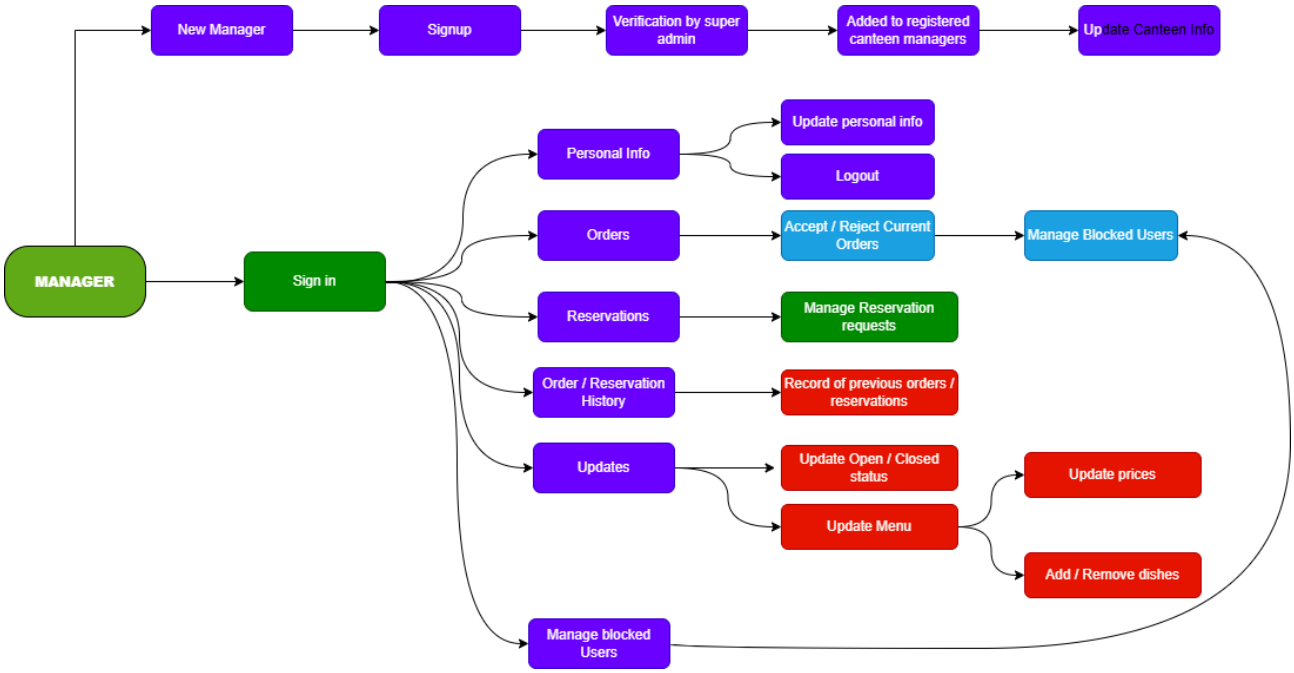


Figure 7: Manager Use Case 2

Author – Velpula Jashwanth

Purpose - The purpose of this use case is to allow a canteen manager to register, manage personal and canteen details, handle orders and reservations, update menu items, and control canteen operations through the system.

Requirements Traceability –

- New manager can sign up
- Manager account is verified by super admin
- Manager can sign in and log out
- Manager can update personal information
- Manager can view and manage orders
- Manager can accept or reject current orders
- Manager can manage reservation requests
- Manager can view order and reservation history
- Manager can update canteen open/closed status
- Manager can update menu and prices
- Manager can add or remove dishes
- Manager can manage blocked users

Priority – High
This use case is very important for smooth canteen operation and system control.

Preconditions –

- Manager must be registered in the system
- Manager account must be approved by super admin
- Manager must be logged into the system
- Internet connection must be available

Post conditions –

- Manager details are saved or updated
- Orders and reservations are processed
- Menu and prices are updated successfully
- Canteen status is updated
- Changes are reflected to users in real time

Actors –

- **Primary Actor:** Canteen Manager
- **Secondary Actors:**
    - Super Admin
    - System (Canteen Management Module)

Exceptions –

- Manager verification rejected by super admin
- System error while updating menu or prices
- Order or reservation processing failure
- Unauthorized access attempt

Includes (other use case IDs): 3.5.1 Admin must approve the Manager for registration

Notes/Issues –

- Only verified managers should access canteen controls
- Menu updates should reflect immediately to users
- System should log all changes made by manager

3.5 Admin Use Case Model

3.5.1 Admin Use Case #1:

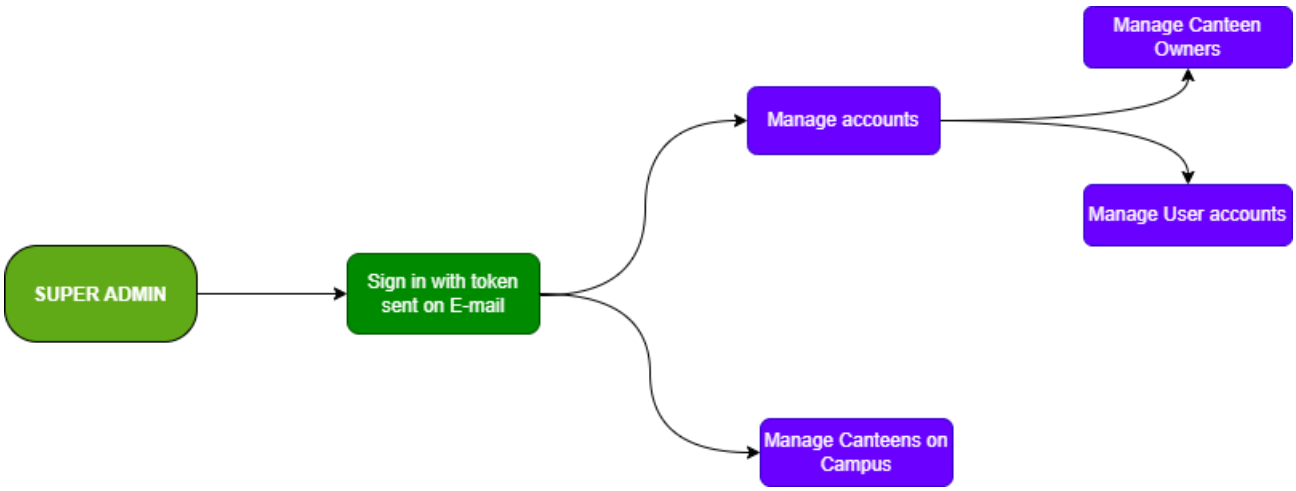


Figure 8: Amin Use Case 1

Author – Boosaraju Sachin

Purpose - The purpose of this use case is to allow the super admin to manage the overall system, including canteens, canteen owners, and user accounts. The super admin ensures the system runs smoothly and securely.

Requirements Traceability –

- Super admin can sign in using a token sent by email
- Super admin can manage system accounts
- Super admin can manage canteen owners
- Super admin can manage user accounts
- Super admin can manage canteens on campus

Priority – High
This use case is critical for system control, security, and administration.

Preconditions –

- Super admin must be registered in the system
- Valid login token must be received via email
- Internet connection must be available

Post conditions –

- Admin is successfully logged into the system
- Canteen and user accounts are managed properly
- System data is updated and saved securely

Actors –

- **Primary Actor:** Super Admin
- **Secondary Actors:**
  - System (Administration Module)

Exceptions –

- Invalid or expired login token
- Unauthorized access attempt
- System error while managing accounts or canteens

Includes (other use case IDs): 3.3.3 & 3.4.2 - Admin can approve/disable accounts of studentss and merchants.

Notes/Issues –

- Login tokens should expire after a fixed time
- All admin actions should be logged for security
- Only authorized super admins should have access

4 Other Non-functional Requirements

4.1 Performance Requirements

4.1.1 User Interactions Response Time

- **Requirement:** The system shall complete user interactions including order placement, reservation booking, and payment processing within a maximum of 3 seconds under normal hours.
- **Rationale:** College students operate on tight schedules between classes and expect rapid transaction completion. Response delays exceeding 3 seconds can result in user dissatisfaction, and reduced platform usage during peak hours.

4.1.2 Database Query and Interface Update Speed

- **Requirement:** The system shall execute standard database operations including menu retrieval, cart modifications, and status updates within 500 milliseconds.
- **Rationale:** Fast responses make the app pleasant to use. When updates take longer than 500 milliseconds, users can feel the delay and the app feels slow.

4.1.3 Peak Load Concurrent User Capacity

- **Requirement:** The system should work smoothly and quickly even when at least 1500 users are using it at the same time during busy hours.
- **Rationale:** College canteens experience concentrated demand during evening times.

4.1.4 Order Processing Throughput

- **Requirement:** The system shall successfully process a minimum of 150 complete order transactions per minute during maximum demand periods.
- **Rationale:** The system must be able to handle many orders at the same time so the canteen can prepare and deliver food without delays or system slowdowns.

4.1.5 API Response Time

- **Requirement:** The system's RESTful API endpoints shall respond to requests within 200 milliseconds for read operations and 400 milliseconds for write operations under normal conditions.
- **Rationale:** Backend API performance directly impacts frontend responsiveness. Optimized Express.js routing, efficient MongoDB queries, and proper indexing ensure rapid data retrieval and updates

4.2 Safety and Security Requirements

4.2.1 Authentication and Authorization

Requirement:
The system shall use secure login with enrollment number/email and password, hashed using bcrypt. Session handling shall use JWT with role-based access. Administrative users must use two-factor authentication.
Rationale:
Strong authentication prevents unauthorized access and protects sensitive system functions.
Safeguards:

- Password complexity rules enforced
- JWT tokens with role data and limited expiry

CIA Triad:

- Confidentiality: Secure authentication ensures that only authorized users can access personal data and protected system features, preventing unauthorized information exposure.

4.2.2 Personal and Financial Data Protection

Requirement:
All personal and payment-related data shall be protected using encryption, secure storage, and access control mechanisms.
Rationale:
Protecting user privacy builds trust and prevents misuse of sensitive information.
Safeguards:

- Encrypted storage of sensitive fields
- Secrets stored in environment variables
- Filtered API responses to avoid data leakage

CIA Triad:

- Confidentiality: Encryption and access controls ensure that personal and financial data is visible only to authorized users.
- Integrity: Secure storage and controlled data handling ensure that stored information remains

accurate and is not modified without permission.

4.2.3 Payment Transaction Security

Requirement:
Payments shall be processed only through certified third-party gateways (Ex – stripe, RazorPay, etc.). The system shall not store card details.

Rationale:
Ensures secure financial transactions and legal compliance.
Safeguards:

- Tokenized payment handling
- Transaction validation on server
- Automatic cancellation of incomplete payments

CIA Triad:

- Confidentiality: Using trusted payment gateways and tokenization ensures that card and payment details are never exposed or stored in the system.
- Integrity: Server-side validation and payment verification ensure that transaction data is accurate and cannot be tampered with

4.2.4 Data Backup and recovery Requirement

Requirement:
The system shall maintain automated backups and disaster recovery procedures for all critical data.
Rationale:
Ensures data availability in case of system failures or attacks.
Safeguards:

- Daily and incremental backups
- Regular recovery testing

CIA Triad:

- Availability – Ensures that data and systems can be quickly restored from backups.

4.2.5 Session and Token Security

Requirement:
User sessions shall be managed securely using JWT with controlled expiration and refresh mechanisms.
Rationale:
Prevents session hijacking while allowing persistent login.
Safeguards:

- Tokens stored in secure cookies
- Short-lived access tokens with refresh rotation
- Token revocation on logout.

CIA Triad:

- Confidentiality – Keeps user sessions private and protected.
- Integrity – Prevents attackers from changing session data.

4.3 Software Quality Attributes

4.3.1 Usability

The user interface shall be simple and clear so students can place an order quickly. Core actions (browse menu, add to cart, confirm order) should be easy to find and complete in a few taps. The website shall give clear feedback (loading indicators, success/error messages) so users know what is happening. The design must work well on phones and laptops.

4.3.2 Maintainability

The codebase shall be organised into small, focused modules so developers can fix bugs or add features easily. All-important code shall have short, up-to-date documentation and automated tests. Version control and a consistent coding style shall be used to help team collaboration.

4.3.3 Interoperability

The system shall communicate using common formats so other services can connect easily (for example, JSON-based APIs). It shall work on major browsers and support common payment or login services if needed. Well-defined API endpoints and clear error messages shall make integrations straightforward. The system should allow other campus systems to read order or menu data when authorised.

4.3.4 Robustness

The application shall handle bad input and dropped network connections without crashing. It shall validate inputs on both client and server and return helpful error messages to users. Automatic retries and safe fallbacks shall be used for temporary failures (e.g., retry a payment check once).

4.3.5 Flexibility

The system shall be easy to change when new features are required (new menu types, outlets, or payment methods). Configuration-driven settings (not hardcoded) shall allow admins to update options without code changes. The design shall support adding new modules with minimal impact on existing features. This lets the service adapt quickly to student needs.

5 Other Requirements

5.1 Legal Requirements and Copyright

User information collected by the canteen digitalization platform will be handled according to government data protection and privacy policies. The ownership and copyright of the system's source code and documentation will be retained by the developers of the project.

5.2 Authentication

OTP verification will be required during the initial registration process. After successful registration, users can create a password which can be used for future logins and accessing the website.

5.3 Data Backup and Recovery

Regular backup of order and user data will be maintained Data recovery mechanisms will be available in case of system failure.

Appendix A – Data Dictionary

A.1 User Database

Variable	Description	Requirements
ID	Unique ID of the user	Unique for each user
Name	Personal information of the user	String
Password	Password of the user	Encrypted string
Email	Email of the user	IITK Email

A.2 Canteen Database

Variable	Description	Requirements
ID	Unique ID of the canteen	Unique for each canteen
Name	Name of the canteen	String
Owner ID	ID of Owner managing the canteen	Unique for each owner
Password	Canteen login password	Encrypted string
Opening Time	Opening time	Time Format (HH:MM)
Closing Time	Closing time	Time Format (HH:MM)
Contact Details	Contact information	String

A.3 Admin Database

Attribute	Description	Requirements
Admin ID	Unique ID of the admin	Unique for each admin
Name	Name of the admin	String
Email	Admin Email Adress	Official Email
Admin Token	Admin Login Code	Encrypted String
Contact Number	Admin Contact number	Numeric
Last Login	Last login timestamp	Date & Time
Status	Admin account status	Boolean

A.4 Dishes Database

Variable	Description	Requirements
ID	Unique ID of dish	Unique for each dish
Canteen Name	Name of the canteen ordering the dish	String
Canteen ID	ID of manager who manages the dish	Unique ID
Menu	List of dishes available in the canteen	List of dishes
Price	Price of the dish	Numeric
Availability	Dish availability	Boolean
Photo	Dish image	File/Image
Other Details	Additional information about dish	String

A.5 Payments Database

Attribute	Description	Requirements
ID	Payment ID	Unique
User ID	ID of User making payment	Valid ID
Amount	Amount of the Payment	Integer
Time	Time at which payment is made	Time
Payment Method	Method used for payment	String
Status	Whether payment was successful or not	Boolean

A.6 Order Database

Attribute	Description	Requirements
ID	Order ID	Unique
User ID	ID of the user who placed the order	Valid ID
Order	Current orders placed	List of dishes
Dish–Quantity	List of dishes and their respective quantities	Pairs
Booking Time	Time at which order was booked	Time
Queue	Current order Queue	List of dishes
Order History	List of past Orders	List of orders

Appendix B - Group Log

Meeting notes and activities will be maintained here

S. No	Date	Timings	Venue	Agenda
1	06/01/2026	20:00–22:00	RM Building	Discussed and proposed various project ideas, including: Canteen Management System, Alumni interaction and student performance tracker.
2	08/01/2026	10:00–12:00	RM Building	Finalized the project idea and brainstormed specific requirements and functionalities.
3	09/01/2026	22:00–23:00	RM Building	Met with the professor to present the project idea and seek suggestions on potential technologies to use.
4	15/01/2026	21:00–22:30	DOMS Building	Reviewed and discussed the SRS (Software Requirements Specification) template.
5	17/01/2026	14:00-14:30	Hall of residence-14	Consulted the Teaching Assistant (TA), divided tasks for completing the SRS document, and established work distribution.
6	22/01/2026	21:00–23:00	P. K. Kelkar Library	Completed the work on SRS document and re-viewed all the work.